

DATA ENGINEERING AND AI

PROJECT PHASE 2

LEARNING CLASSIFYER SYSTEMS FOR CUSTOM CHURN PREDICTION

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Task 1: Project Phase – I Summery

Project Phase – I Summary:

The selected problem of this project was the customer churn prediction in the telecommunication industry. A churn customer occurs when customers stop using a company's services and instead switch to another company or provider. Predicting churn is very important because the existing customers are generally more cost effective than switching companies. The objective of this project was to develop and evaluate the Learning Classifier System (LCS) be capable of learning and predicting whether a customer is likely to churn based on the past customer information.

The dataset used for this project was the Telecom Customer Churn Dataset, which contains information relating to customer demographics, subscription services, account details, service usage and their billing behaviours. The Dataset consists of 7,043 customer records and 38 original features. A binary target variable called the Churn target was used for the prediction, where 1 represented a churned customer and 0 represented a customer who stayed with the provider. The dataset contained of many mixtures of numerical and categorial variables, providing a comprehensive representation of customer characteristics that may have an influence on the churn behaviour. Customers who have joined the company will not be considered and excluded as they are irrelevant to the question we initially posed for this project.

During Phase I, a few data-quality issues were identified. Missing values were present in multiple service-related attributes and the whole project required thorough investigation before starting the modelling. The dataset also contained numerous categorial variables that had to be transformed into numerical variables for the machine learning models. Class imbalance was observed, with fewer churned customers than the retained customers. Several attributes, including Customer Status, churn category and Churn Reason were identified as the potential target- leakage variables because they contained information that would not be available when predicting future churn. Also, several records contained negative monthly charge values, which required further investigation to determine whether they represented valid data or not.

Customer churn prediction supported important business decisions within the telecommunications company. By identifying the customers who were at high risk of leaving, the company could implement retention strategies such as targeted promotions, service plans, or customer support interventions, and more. This can help reduce the customer loss, improve customer satisfaction and increase long term revenue.

A learning Classifier System (LCS) was selected as the main modelling technique because the customer churn prediction is a supervised classification problem. LCS models are capable to learn both numerical and categorial data while generating interpretable rule-based predictions. Unlike any other traditional machine learning algorithms that operates as black-box models, LCS provides human-readable rules that explain how the predictions are or could be made. This interpretability allows the business stakeholders to understand the factors associated with the customer churn and supports the development of a more effective and efficient customer

retention strategies. Therefore, the customer churn prediction problem was considered highly suitable for an LCS- based solution.

Task 2: Original eLCS System on raw Data sheet

Objective:

The original eLCS model was developed to provide the starting point of the project and establish a baseline for customer churn prediction. The baseline allowed the performance of the model to be measured before the preprocessing, feature engineering and system improvements were applied. It was possible to better compare the later results and determine whether the changes made throughout the project improved the model's ability to predict the customer churn. Overall, the baseline model served as a foundation for the entire project, providing a reliable starting point from which the impact of each improvement could be measured and assessed.

Data Preparation:

The dataset was prepared before training the baseline model to ensure the compatibility of the eLCS algorithm. The churn target was introduced and used, where the customers who churned were assigned a value of 1 and the customers who remained within the company were assigned the value of 0. The potential leakage variables, including the Customer Status, Category and Churn Reason were removed because they contained information that would not be available while predicting the future churn behaviour. Including these values would produce unrealistically high model performance and reduce the validity of the evaluation. The customer IDs were also excluded because it does not provide the predictive value.

Train – Test Split:

The dataset was divided into separate training and testing subsets using an 80/20 stratified train-test split to ensure that the subsets were evaluated fairly and objectively. This meant that 80% of the data was used to train the model and the remaining 20% was reserved for testing its performance. A stratified split was chosen to maintain the original proportion of the churned and the non-churned customers in both the datasets because the dataset contained an imbalanced class distribution. Hence, preserving this distribution helped ensure that both the training and testing datasets accurately represented the characteristics of the original dataset. Also, a random state of 42 was applied to ensure reproducibility of the results. Separating data into training and testing sets also reduce the risk of overfitting. If the model was evaluated using the same data on which it was trained, the performance results could be overly optimistic and may not reflect its true predictive capability. By using unseen test data, a more reliable and accurate measure of the model's generalisation performance was obtained.

Original eLCS Configuration and its parameters:

The original eLCS was used as the baseline model without introducing any additional enhancements. This allowed a fair comparison between the baseline system and the improved eLCS model developed later in the project. The model was trained using the default eLCS parameters provided by the implementation. These parameters controlled important aspects of the rule population, learning process and generic algorithm operations. The baseline configuration was left unchanged to provide a benchmark for evaluating later improvements.

Parameter	Value
Population size (N)	1000
Learning iterations	10,000
Learning Rate (Beta)	0.2
Crossover Probability (Chi1)	0.8
Mutation rate (Mu)	0.04
Accuracy Threshold (Acc_sub)	0.99
Genetic Algorithm Threshold (Theta_GA)	25
Deletion Threshold (Theta_del)	20
Subsumption Threshold (Theta_sub)	20
Selection Threshold (Theta_sel)	0.5
Prediction Specificity Probability (P_spec)	0.5
Selection Method	Tournament
Initial Fitness	0.01
Fitness Reduction	0.1
Power Parameter (Nu)	5
Deletion Fraction (Delta)	0.1
Genetic Algorithm Subsumption	Enabled
Correct Set Subsumption	Enabled

Model Training and Evaluation:

After the training data is prepared, the original eLCS model was trained to learn the classifications rules that distinguish the churned customers from the retained customers. Predictions were made by using and evaluating datasets using several classification metrics. The evaluation metrics included Accuracy, Precision, Recall, F1 Score, and Balanced Accuracy. These metrics are important for the evaluation of the model performance from multiple perspectives rather than relying only on accuracy.

Baseline Results:

The original eLCS baseline model achieved an accuracy of 80.70%, showing that it was able to correctly classify most customer records in the testing dataset. The model achieved a precision score of 69.62%, meaning that many customers predicted to churn were actual churners. However, the recall score of 48.40% indicates that the model missed a noticeable number of customers who eventually churned. As a result, the model achieved an F1 score of 57.10%.

The balanced accuracy of 70.38% suggests that the model performed reasonably well despite the class imbalance in the dataset. These results show that the baseline eLCS was proved to be more helpful and useful at identifying non-churning customers than detecting customers who are likely to churn. The relatively low recall indicates that several churns were missed which highlights the opportunity for improvement. Overall, the baseline model provided a solid starting point for the project, but its lower recall highlighted the need for further preprocessing, feature engineering, and model improvements to better identify customers at risk of churning.

Metric	Value
Accuracy	0.806955
Precision	0.696154
Recall	0.483957
F1 Score	0.570978
Balanced Accuracy	0.703814

Task 3: Data Preprocessing and Feature Engineering

Overview:

Several Preprocessing and Feature engineering techniques were applied before developing the improved eLCS model to improve the data quality and prepare the dataset for machine learning. The preprocessing stage was focused on handling missing values, encoding categorical variables, correcting invalid data, addressing class imbalance, and selecting relevant features for the churn prediction.

Data Leakage Prevention:

The potential data leakage variable was removed before model training to ensure a fair evaluation. The attributes like Churn category, Customer Status and Churn Reason contained the information related to churn customers and would not be available while predicting the future customer behaviour. If these variables were included, they would have caused and produced a misleading performance result. Also, the customer IDs were also removed because it served as a unique identifier and provided no predictive value.

Missing Value Treatment:

Several attributes had missing values that required several and deep investigations. Many of these values occurred since customers had not subscribed to certain services. To preserve the data, missing values had to be replaced with other meaningful categories. Numerical service-related attributes were also replaced with zero wherever appropriate.

Invalid Value Investigation:

During data exploration, negative values were identified within the monthly charge attribute. Since most monthly service charges cannot realistically be negative, the values were classified to be invalid. Around the total of 120 records contained negative charges, 97 records in the training dataset and 23 records in the testing dataset were negative monthly charges. These values were all replaced using the median calculated from the training data to avoid data leakage.

Feature Selection:

The feature selection was performed by Mutual Information (MI) analysis to identify the attributes that are strongly associated with the customer churn. High MI scores were considered

more informative and were retained for the use in the improved eLCS model. This process reduced the noise within the dataset and allowed the model to focus on the most relevant customer characteristics. By removing the less informative feature, the complexity of the dataset would reduce, making it easier for the model to identify meaningful patterns. Another advantage of Feature selection is that it improves Interpretability. This allows important factors that influenced and continues to influence churn customers to be identified more clearly and helped provide valuable insights for all the business decision making.

Categorical Encoding and Class Imbalance handling:

The machine learning algorithms cannot directly process text-based categories. Hence, categorical features were converted into numerical representations using ordinal encoding. There was a class imbalance present in the dataset as well with considerably fewer churned customers than retained customers. There were 3139 non-churn customers and 1495 churn customers in the original training distribution. To reduce the model bias towards the majority class, random oversampling was applied to the minority case., this produced a balanced training dataset containing 4139 records in each case.

Preprocessing Summary:

The preprocessing stage played an important role in improving the overall quality and usability of the dataset before model development. Several data quality issues identified during Phase I were addressed, including target leakage, missing values, invalid data, class imbalance, and categorical attributes. By removing leakage variables, the model was prevented from learning information that would not be available in real-world prediction scenarios.

Missing values were treated according to their context, invalid monthly charge values were corrected, and categorical features were converted into numerical representations suitable for machine learning. In addition, random oversampling was applied to reduce the impact of class imbalance, while feature selection helped identify the attributes that were most relevant to customer churn.

Overall, these preprocessing and feature engineering steps created a cleaner, more consistent, and more reliable dataset. This provided a stronger foundation for model training and helped improve the ability of the improved eLCS model to learn meaningful patterns related to customer churn.

Task 4: Improved LCS – Based System

Objective:

The objective of the improved eLCS System was to enhance the performance of the original baseline model by processing preprocessing techniques, feature engineering and optimisation. The improved system was based to build upon the baseline model parameters to strengthen the learning process. These improvements were designed to help the model learn more meaningful relationships between customer behaviour and churn behaviour.

Improvements implemented:

There were several improvements that were implemented.

The improved eLCS model was trained using the preprocessed dataset developed in Task 3. Before training, the data was cleaned and prepared through several preprocessing steps, including handling missing values, correcting invalid monthly charge values, encoding categorical attributes, balancing the class distribution, and selecting the most relevant features. These steps helped improve the overall quality of the dataset and ensured that the model learned from more meaningful and reliable information. In addition to improving the dataset, several eLCS parameters were adjusted to strengthen the learning process. The number of learning iterations was increased from 10,000 to 15,000, giving the model more opportunities to learn, refine, and optimise its classification rules. The maximum classifier population size (N) was also increased from 1,000 to 2,000, allowing the system to maintain a larger and more diverse collection of rules. These changes were introduced to improve the model's ability to capture complex patterns in customer behaviour and generate more effective predictions.

Improved Training process:

The improved eLCS model followed a structured machine learning process. Once the preprocessing stage was completed, the cleaned, balanced, and encoded training dataset was used to train the model. The revised eLCS parameter settings were then applied to help improve the model's learning performance. During training, the model learned a set of classification rules based on patterns found in the customer data. These rules were then used to make predictions on the testing dataset, which had not been seen by the model before. To evaluate performance, the same metrics used for the baseline model were applied, including accuracy, precision, recall, F1 score, and balanced accuracy. Using the same evaluation criteria ensured a fair comparison between the original and improved eLCS models and made it easier to assess the impact of the improvements introduced throughout the project.

Discussion:

The results show that the preprocessing techniques and parameter changes had a positive impact on the model's ability to predict customer churn. While the improved eLCS achieved a slightly lower accuracy than the baseline model, it produced a much higher recall score. This means the model was able to identify more customers who were likely to churn, which was one of the main goals of the project. In customer churn prediction, it is often more important to identify potential churners than to maximise overall accuracy, as missed churners represent lost opportunities for customer retention. The higher recall achieved by the improved eLCS suggests that the model would be more useful in a real-world telecommunications setting where retaining at-risk customers is a priority. Overall, the results demonstrate that data preprocessing, feature engineering, and parameter optimisation helped improve the model's effectiveness. The improved eLCS was better able to learn meaningful churn-related patterns and provided a stronger basis for comparison with conventional machine learning models.

Task 5: Experimental Design and Evaluation

We decided to complete an 80/20 train-test split for our validation strategy approach. We reasoned that since we have a rather substantial dataset in records, the 80% provided to a model would be more than sufficient in providing us with a decent result given plenty of records, while also leaving plenty in the 20% for evaluation. To that the k-fold would be excessive compared to the 80/20 approach when we have such a large record population. We were aware that the 80/20 approach produces different values between training and test sets on a class divided database like ours, so to combat this problem we enabled stratification so that it would reduce the problem.

The notebook has calculated the fundamental metrics when ran by the Improved eLCS model

Firstly, we observe a 76.86% accuracy metric. This metric measures the percentage of the population that the model correctly classified, this also means of the entire population of customers the Improved eLCS model had incorrectly classified 23.14% of our population. It should be noted that this metric should be taken cautiously at face value, the portion of the customer population that Stayed is significant and the majority. This means the model could therefore get a reasonably high accuracy metric while actually being poor at correctly classifying churning customers.

Next, the precision metric came out to be 54.15%. This metric measures the percentage of predictions done by the Improved eLCS model that are actually correct. This equates to roughly half of all predictions done by the Improved eLCS model to be correct, while the rest are false positives. We should be careful using this metric or this model for the actual implementation of this project because approximately 45.85% of customers predicted to churn by this model were non churners, displaying a relatively high rate of false positive churn predictions.

Next, recall came in at 83.69%. This metric put in context measures of all the customers who churned, how many in percentage did the model successfully identify. This also means 16.31% of all churners were not correctly identified by the Improved eLCS model and had slipped through correct identification.

The F1 score is a metric that combines both precision and recall. We observe a 65.76% score for the F1 metric. This metric is instrumental in a model to measure both correctly identifying churners as well as avoiding false positives, both of which are valuable for the application of this project for the telecommunications company.

Balanced accuracy came out to be 79.04%. This metric measures the per-capita accuracy of both classes churned and stayed and is especially useful in a situation like this where we have a majority class in stayed which can tip and mislead the accuracy metric.

Please note ROC-AUC and PR-AUC values were calculated, but currently they're not present for the various other models, thus we reasoned that it would be best to not include them for the time being to avoid an inconsistent comparison.

We also selected and completed a McNemar's statistical test for the Improved eLCS model vs the Random Forest model. We choose this test because McNemar's test is designed for use on paired nominal outcomes which matches our situation exactly where we have Improved eLCS

and Random Forest paired on predicting the same 1,409 test customers on an outcome either right or wrong.

McNemar's test results showed Improved eLCS got 73 counts only correct, while Random Forest got 168 counts only correct. This was of 241 discordant predictions total, generating us a McNemar p-value of $8.559183 \cdot 10^{-10}$. This result is significantly below the conventional significance value of 0.05, therefore we have sufficient evidence to reject the null hypothesis being tested by the McNemar's test being; 'among cases of two classifiers disagreeing, neither classifier model is systematically more likely to be correct than the other'.

In other words, we have strong evidence that there is a difference in the paired classification performance between Improved eLCS and Random Forest. It should be noted that this should not be interpreted as 'Random Forest is therefore the better model'.

Task 6: Comparison with Conventional Machine learning models

For our model comparisons we have chosen three conventional non-deep-learning machine learning models to compare against each other. Our choices were; Logistic Regression model, Decision Tree model, and Random Forest model. These three models were trained and evaluated on the same test split that was used by the Improved eLCS so a fair comparison could be expected.

We choose these three models in particular as they embody definitively different modelling approaches, and we reasoned such models would give us substantially different results as opposed to three models that behaved similarly to which comparison would be most valuable in. Logistic Regression provides us with a simple linear baseline good for binary prediction, Decision Tree approaches non-linearly and is rule based, and Random Forest is an ensembled tree-based model (ensemble meaning this one model is combining many models) which uses many decision trees.

We also ran the Original eLCS model on both minimally processed data and preprocessed data. However, it must be noted that the minimally processed original eLCS ran on 1,318 test cases whereas the others ran on 1,409. It would be inaccurate to say the test is completely fair. The results when comparing other models to the minimally processed original eLCS should be taken with caution.

Compiled table of various metrics for each machine learning model ran:

Model	Accuracy	Precision	Recall	F1	Balanced Accuracy
<i>Logistic Regression</i>	77.79%	55.21%	86.36%	67.36%	80.52%
<i>Decision Tree</i>	76.15%	55.19%	54.01%	54.59%	69.08%
<i>Random Forest</i>	83.61%	70.37%	66.04%	68.14%	78.00%
<i>Improved eLCS</i>	76.86%	54.15%	83.69%	65.76%	79.04%
<i>Original eLCS (minimally processed)</i>	81.26%	71.53%	56.42%	63.08%	73.76%
<i>Original eLCS (Preprocessed)</i>	76.79%	54.41%	77.54%	63.95%	77.03%

Comparing between original eLCS on minimally processed, preprocessed, and Improved eLCS we observe a clear trend; Preprocessing seems to have changed the behaviour of eLCS significantly.

We observe recall seems to have significantly increased from 56.42% in minimally processed to 77.54% in preprocessed, then increased even further to 83.69% with modifications.

The F1 metric value also seems to have slightly increased with processing and modifications slightly rising from 63.08% to 63.95% to 65.76%.

Balanced accuracy has a similar effect though slightly larger than F1, going from 73.76% in minimally processed to 77.03% to 79.04%

Accuracy drastically drops between minimally processed to preprocessed then fluctuates up very slightly when modified, from 81.26% to 76.79% then lastly to 76.86%.

Precision also shows a clear downwards trend as processing and modifications are implemented. Going from a modest 71.53% to a huge decline to 54.41% then to 54.15%.

The interpretation that we can draw from such results say that as preprocessing and LCS modifications are implemented, it has shifted the model towards a better ability at detecting more actual churning customers but does so at the expense of a slightly lowered accuracy score as well as a major hit to precision.

Next, we can analyse the comparison between our proposed Improved eLCS and the other conventional non-deep-learning machine learning models starting with Logistic Regression.

The first trend we can identify is that the differences between the performance metrics of both models are miniscule in comparison to what we have seen so far. The differences in metric percentages are often only varying by a few points at most. This tells us that Improved eLCS and Logistic Regression performs near identically when judged by just these metrics. However, we can also notice that in the metrics measured; accuracy, precision, recall, F1, and balanced accuracy Logistic Regression is always performing higher, it wins over Improved eLCS in every metric, though as said previously the differences are minor. One small thing to note Logistic Regression scores the highest recall metric value out of our six models, if the company wanted a model that could identify as many potential churners as possible out of a database, then Logistic Regression would be their best choice.

Next, for Random Forest vs Improved eLCS we can see that Random Forest boasts a significantly higher precision value (54.15% vs 70.37%) while also sporting a meaningfully higher accuracy value (76.86% vs 83.61%). This suggests that Random Forest is much better at avoiding false churn predictions and better population identification than Improved eLCS. However, it comes at the cost of a significantly lower Recall value in comparison (83.69% vs 66.04%) meaning we can expect Improved eLCS to catch significantly more actual churners than Random Forest could, 17.65 percentage points is a significant difference and will definitely matter depending on what the telecommunications company's goals are for such machine learning models. But it definitely shows that it's rare for a model to be significantly superior to another on all fronts but rather is best to choose one based on specific application or goal in mind.

Lastly, the difference in performance metrics between Decision Tree vs Improved eLCS is uneventful. They both benchmark roughly the same scores for Accuracy and Precision with Improved eLCS being slightly better than Decision Tree with a less than 1% percentage point difference, and Decision Tree beating Improved eLCS with a slightly higher Precision metric with a 1.04% percentage point difference. A more significant difference appears in Recall when

comparing the two models, Improved eLCS does significantly better with an 83.69% recall value when compared to Decision Tree's 54.01%. As for F1 and Balanced Accuracy Improved eLCS generally does better with a roughly ~10% higher percentage point difference in both metrics. Furthermore, the F1 metric of Decision Tree compared to all other models is substantially lower than all others. Overall, Decision Tree ranks as easily the weakest amongst all other models tested, it doesn't do substantially well in any of its better metrics but heavily falls behind in its weaker metrics.

Task 7: Rule Extraction and Interpretation

The model comparison we completed showed that no model was definitively the best throughout all metrics. For example, the Improved eLCS model achieved a high recall value, Random Forest achieved the highest accuracy, precision, and F1 score of all our models. Likewise, the logistic Regression model achieved the highest recall and balanced accuracy value. Therefore, the most 'suitable' model depends on the intended purpose of said model.

Our files extracted 1,903 total learned rules and 212 experienced churn rules.

The three top rules that were sufficiently experienced and high quality are examined:

Rule 1 had a 100% accuracy across 23 instances tested and used features including Dependents, Married, Age, Streaming TV, Avg Monthly Long Distance Charges, Streaming Movies, Internet Service and Contract.

Rule 2, 98.21% accuracy across 56 instances, using features such as: Number of Dependents, Streaming Movies, Contract, Payment Method, Number of Referrals, Online Security and Total Revenue.

Rule 3, 97.87% accuracy across 94 instances on features such as: Number of Referrals, Payment Method, Phone Service, Contract, Paperless Billing, Married, Number of Dependents, Total Refunds and Avg Monthly Long Distance Charges.

These top rules along with the others demonstrate the interpretability advantage to using LCS. We can represent the model's decisions through a combination of such feature conditions.

However, again in statistics certain values should be taken carefully and not at face value. This is the same, the rules found should not be considered as proof that these features are causations of churn. They simply represent the exact patterns the models have learned by being trained on such data. Rules should be used to guide, to suggest patterns, and as evidence, but should not be interpreted as proof of causation.

Task 8: Discussion, Limitations, and Future Work

Lastly for this investigation, we are going to critically discuss the results we found through our investigation and the responsible use of the system.

We have found the Improved eLCS model to have achieved a 76.86% accuracy, 54.15% precision, 83.69% recall, 65.76% F1, and 79.04% balanced accuracy.

As we can see from these metrics our Improved eLCS model is best applied as a customer churn detection model as indicated by the significantly high recall performance metric. The high recall score means that the model is able to correctly identify a large proportion of customers who actually churned.

However, as we have been shown throughout the investigation there is rarely a model that is vastly superior to another on all fronts and metrics that is the definitive best choice when it comes to all applications and uses.

Critically, of all our models and tests; Random Forest achieved the highest accuracy, precision, and F1 score. Logistic Regression achieved the highest recall and balanced accuracy score.

Therefore, although our Improved eLCS model was competitive it clearly falls short when it comes to certain performance metrics when compared to other models.

So, it stands to reason that the definition of a 'best' or 'most suitable' model changes depending on the goals and applications the telecommunications company decides is best aligned with their needs. Therefore, depending on the purpose of the model the company decides is best, the value and worth of each model changes. It would be misleading to say there is a best overall model, but we can say there are models more suited for each purpose that the company decides they wish to pursue. One model may be significantly 'better' than another but only for the exact purpose the company chooses for it.

Furthermore, the changes we made during preprocessing also had a noticeable effect on the metrics and behaviour of the eLCS model. We observed both increases and decreases in certain metrics as preprocessing is implemented indicating the growth of both strengths and weaknesses. Therefore, data processing should be done cautiously as it may affect the performance of the model that relies on it.

Next, the McNemar test we did comparing the Improved eLCS model and the Random Forest model showed a statistically significant difference in the outcomes. The p-value generated (8.56×10^{-10}) is significantly below the 0.05 value we commonly used as a threshold, which means we have strong evidence to say that among the two models neither model is systematically more likely to be correct than the other. Further conveying that the most appropriate model depends on the uses intended for it.

It should be noted that there are several limitations to our investigation.

Firstly, the original eLCS model ran on minimally processed data was trained on 1,318 test records while the other models were trained on 1,409 records. While the difference is small, it

does mean that the face-value comparison between metrics that uses the values generated should be taken carefully as it may be misleading.

Secondly, we manually selected and choose the parameters to be used with the Improved eLCS model. As a consequence of human reasoning and choice making, we cannot guarantee with certainty that the selected parameters were the most optimal. Therefore, the performance of the Improved eLCS model may vary and should be taken with caution.

We used random oversampling during our preprocessing to make up for the class division present in our database. Although it does address the class imbalance to some degree, it may also introduce and affect the model's metrics, we cannot say with certainty that it would not.

We should also note the responsible use of such models and the dangers associated with taking the results as fact. The generated prediction is simply just a prediction; it is not definitive and not always true. Therefore, results should be used to guide rather than make clear cut decisions. We recommend using it to supplement human decision making if the company decides to implement such techniques. Using it incorrectly may lead to negative effects, for example a worse customer experience at one end of the spectrum and at the other the company loses more money due to incorrectly predicting a customer will churn and thus sending them offers with less profit margins.

There are many future improvements we could do to better the results of this investigation. For example; we could test our findings and models on other telecommunication datasets to better assess generalization and whether it applies to more than just this database, we could bring in more teammates to our investigation and ask their stance on their chosen method of preprocessing to increase the variety and logical reasoning behind our choices, we could seek external validation by exploring other methodologies and techniques done by other investigations that we may learn and employ, and more.

Overall, the Improved eLCS model is a viable model for our needs in customer churn prediction, particularly for its ability in identifying as many potential churners as possible. However, the results we found throughout this study demonstrates the various limitations models may have, the trade-off strengths and weaknesses present in different models, as well as how we use the results generated by our models.